

Erdős Problem #917

EP-917	L1 - Tractable	Partiellement résolu	Graph Theory
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Énoncé original (anglais, reproduction mot pour mot)

Let $k \geq 4$ and $f_k(n)$ be the largest number of edges in a graph on n vertices which has chromatic number k and is critical (i.e. deleting any edge reduces the chromatic number).

Is it true that $f_k(n) \sim k n^2$? Is it true that $f_6(n) \sim n^{2/4}$? More generally, is it true that, for $k \geq 6$, $f_k(n) \sim \frac{1}{2} \left(1 - \frac{1}{\lfloor k/3 \rfloor}\right) n^2$?

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